Healthcare Claims Fraud Detection

Synthetic Business Rulebook — Version 1.0

Illustrative POC rules only; not official payer, medical coding, legal or regulatory guidance.

1. Scope and interpretation

The rules below are intended to create explainable investigation signals in synthetic data. They do not establish fraud. A claim may trigger a rule for legitimate reasons, and an unflagged claim is not guaranteed to be valid.

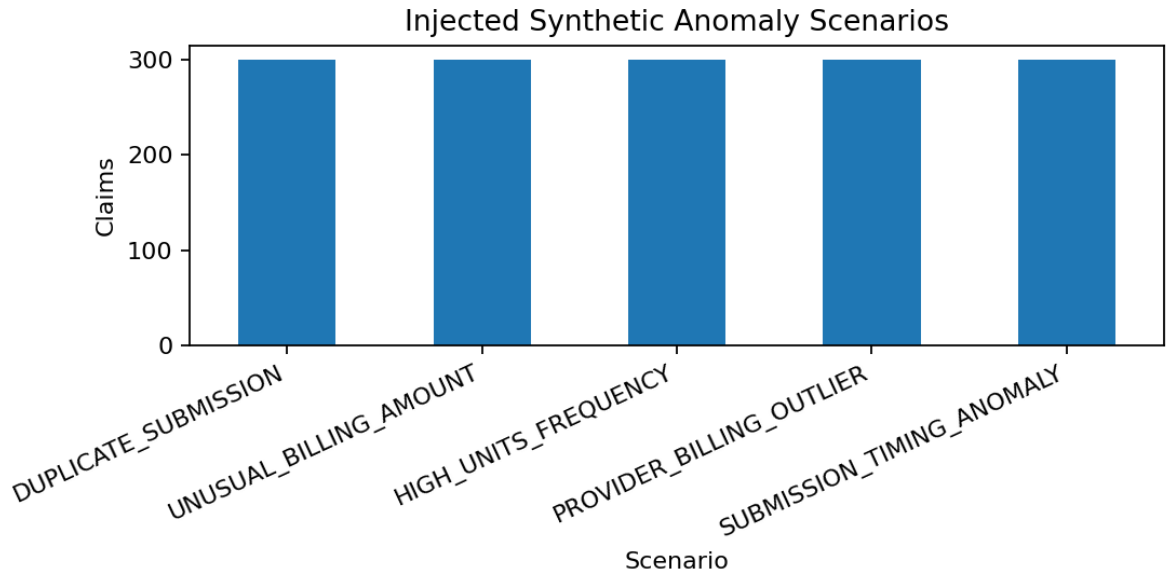
2. Rule catalogue

Rule ID	Rule	Illustrative condition	Severity	Evidence surfaced
HC-DUP-001	Exact/near duplicate	Same member, provider, service date, procedure and units; compare amount and submission timing.	High	Matching claim IDs and matching fields
HC-AMT-001	Unusual billed amount	Billed amount materially above a procedure reference amount or specialty peer distribution.	Medium/High	Amount ratio, peer percentile, reference amount
HC-UNIT-001	High units	Units exceed the synthetic procedure guideline or are extreme versus comparable claims.	Medium/High	Units, guideline and claim amount
HC-PRV-001	Provider billing outlier	Provider aggregate amount or frequency deviates from specialty/region peers.	Medium/High	Provider metrics and peer baseline
HC-TIME-001	Submission timing anomaly	Submission occurs unusually long after service date relative to configured POC window.	Low/Medium	Service date, submission date and lag days
HC-ML-001	Multivariate anomaly	Isolation Forest identifies an unusual combination of claim features.	Review signal	Anomaly score and contributing feature context

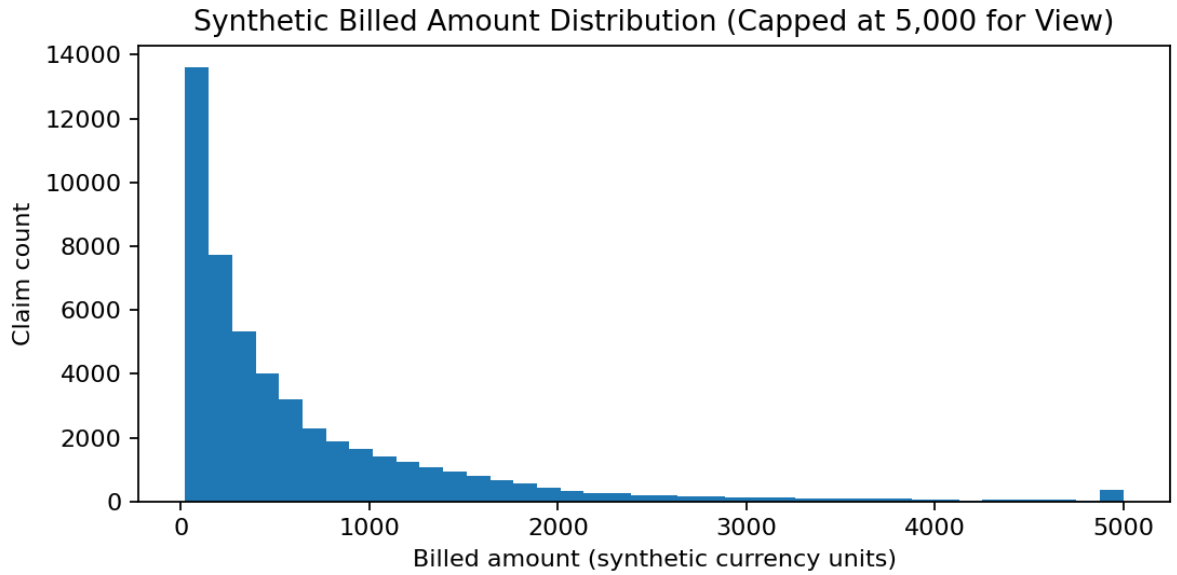
3. Synthetic data profile

The bundled dataset contains 50,000 claims, with 1,500 claims carrying a synthetic anomaly label (3.0%). The labels are created by design for controlled evaluation and are not real-world fraud outcomes.

Injected scenario distribution



Billed amount distribution



4. Rule implementation notes

HC-DUP-001 — Duplicate candidates

Use a composite key from member_id, provider_id, service_date, procedure_code and units. Amount and submission date help distinguish exact from near duplicates. Duplicate status should remain a candidate until reviewed.

HC-AMT-001 — Billing amount deviation

Compare billed_amount against procedure reference_amount and peer-group statistics. Robust measures such as median and median absolute deviation can reduce sensitivity to extreme values. Configure thresholds in a YAML file rather than hard-coding them.

HC-UNIT-001 — High units

Compare units against maximum_units_guideline in procedures.csv. The guideline is synthetic and should be treated as a test fixture only.

HC-PRV-001 — Provider outlier

Aggregate claim count, median billed amount and high-amount share by provider. Compare providers within a similar specialty/region peer group and require a minimum sample size.

HC-TIME-001 — Submission timing

Calculate submission_lag_days = submission_date - service_date. Long delays may have legitimate operational explanations and should not independently imply fraud.

HC-ML-001 — Unsupervised anomaly

Fit Isolation Forest only on claim-derived features. Exclude claim_id, synthetic_anomaly_label and synthetic_scenario from model inputs. Review model scores alongside rule evidence.

5. Risk score and review bands

Initial score weights are: duplicate 25%, amount deviation 20%, upcoding-like indicators 15%, provider peer outlier 15%, frequency 10%, and Isolation Forest signal 15%. These weights are POC assumptions and should be calibrated using held-out synthetic scenarios and reviewer workload.

Score	Display band	Suggested handling
70–100	High	Prioritize for human review; inspect rule evidence and similar claims.
40–69	Medium	Review when capacity permits or when additional evidence exists.
0–39	Low	Lower priority; continue normal controls and sample monitoring.

6. Label separation and evaluation

The synthetic ground-truth label table is stored separately in synthetic_labels.csv. The detection pipeline must score claims without access to these columns. Only after predictions are generated may the labels be joined for evaluation. Report precision, recall, F1, PR-AUC where applicable, false-positive counts, scenario-level detection and review volume. Do not describe synthetic-label performance as real-world performance.

7. Review workflow

1. Open the claim and confirm the member/provider/service attributes. 2. Inspect the triggered rules and comparison baseline. 3. Check duplicate candidates and provider peer metrics. 4. Record a reviewer outcome and reason code. 5. Track false positives and update rule thresholds through documented change control.